Abhinav Kumar

Ph.D. Student | AI Safety | Agentic Systems

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Research Focus

I develop **trustworthy AI systems for real-world deployment**, with a focus on the security, safety and failure modes of generative reasoning models in **agent and multi-agent settings**.

Education

University of Massachusetts Amherst — Sept 2024 – May 2028
Ph.D. in Computer Science — Advisor: Eugene Bagdasarian

Saint Louis University — Aug 2023 – Jul 2024
M.S. in Artificial Intelligence — Advisor: Reza Tourani

Saint Louis University — Aug 2019 – Jul 2023
B.S. in Computer Engineering

Research Experience

Max Planck Institute for Intelligent Systems, ELLIS Institute — Sept 2026 – Present
Guest Researcher

Max Planck Institute for Intelligent Systems, ELLIS Institute — June 2026 – Sept 2026
Visiting Ph.D. Student — Advisor: Sahar Abdelnabi

- Researching safety risks arising from miscoordination in multi-agent systems.

ServiceNow — Feb 2026 – Present
External Collaborator

- Developing benchmarks for contextual integrity in multi-agent systems.

University of Massachusetts Amherst — Sept 2024 – Present
Graduate Research Assistant

- Studying vulnerabilities and defenses for trustworthy AI systems.
- Developing frameworks to benchmark and study agent and multi-agent safety.

Intel Labs — Aug 2021 – Jul 2024
External Collaborator

- Developed frameworks and algorithms for pervasive edge privacy.

Saint Louis University — May 2021 – Jul 2024
Research Assistant

- Developed TEE based frameworks for privacy-preserving machine learning, MLaaS validation, and trustworthy AI systems.
- Conducted research on continual learning backdoors, machine unlearning and distributed learning.

Awards & Grants

BlueDot Rapid Grant — Sept 2026
OpenAI Researcher Access Program — Sept 2025
IEEE/IFIP DSN Best Paper Award — Aug 2025
IEEE PerCom Best Industry Track Paper Award — Mar 2022
Saint Louis University Scholarship — Aug 2019

Publications

Preprints

Terrarium: Revisiting the Blackboard for Multi-Agent Safety, Privacy, and Security Studies. Mason Nakamura*, **Abhinav Kumar***, Saaduddin Mahmud, Sahar Abdelnabi, Shlomo Zilberstein, Eugene Bagdasarian.
arXiv, 2025. (*Equal Contribution)

Throttling Web Agents Using Reasoning Gates. **Abhinav Kumar**, Jaechul Roh, Ali Naseh, Amir Houmansadr, Eugene Bagdasarian.
arXiv, 2025.

OverThink: Slowdown Attacks on Reasoning LLMs. **Abhinav Kumar**, Jaechul Roh, Ali Naseh, Marzena Karpinska, Mohit Iyyer, Amir Houmansadr, Eugene Bagdasarian.
arXiv, 2025.

Peer-Reviewed Conferences

Colosseum: Auditing Collusion in Cooperative Multi-Agent Systems. Mason Nakamura*, **Abhinav Kumar***, Saswat Das*, Sahar Abdelnabi, Saaduddin Mahmud, Ferdinando Fioretto, Shlomo Zilberstein, Eugene Bagdasarian.
NeurIPS 2026. (*Equal Contribution)

PiSAs: Benchmarking Contextual Integrity in Multi-User Agentic Systems. Shubham Gupta*, Nazanin Mohammadi Sepahvand*, **Abhinav Kumar**, Cem Subakan, Spandana Gella, Pierre-André Noël, Perouz Taslakian, Eugene Bagdasarian, Valentina Zantedeschi.
COLM HAIPS Workshop, 2026. (*Equal Contribution)

Network-Level Prompt and Trait Leakage in Local Research Agents. Hyejun Jeong, Mohammadreza Teymoorianfard, **Abhinav Kumar**, Amir Houmansadr, Eugene Bagdasarian.
USENIX Security, 2026; ICML AIWILD Workshop, 2026.

Privacy Analysis of Oblivious DNS over HTTPS: a Website Fingerprinting Study. Mohammad Salari, **Abhinav Kumar**, Federico Rinaudi, Reza Tourani, Alessio Sacco, Flavio Esposito.
IEEE/IFIP DSN, 2025. **BEST PAPER AWARD**

Persistent Backdoor Attacks in Continual Learning. Zhen Guo, **Abhinav Kumar**, Reza Tourani.
USENIX Security, 2025.

Silver Linings in the Shadows: Harnessing Membership Fingerprinting for Machine Unlearning. Nexhi Sula, **Abhinav Kumar**, Jie Hou, Han Wang, Reza Tourani.
IEEE CNS, 2025.

A Generative Framework for Low-Cost Result Validation of Machine Learning-as-a-Service Inference. **Abhinav Kumar**, Miguel A. Guirao Aguilera, Reza Tourani, Satyajayant Misra.
ACM ASIACCS, 2024.

SCLERA: A Framework for Privacy Preserving MLaaS at the Pervasive Edge. **Abhinav Kumar**, Reza Tourani, Mona Vij, Srikathyayani Srikanteswara.
IEEE PerCom, 2022. **BEST INDUSTRY TRACK PAPER AWARD**

Invited Talks

OverThink: Slowdown Attacks on Reasoning LLMs.
Google Adversarial ML Knowledge Sharing, Mar 2025.

DcTC: Towards Data-centric Trusted Computing.
Intel SPR Tech Talks, Jul 2025.

A Generative Framework for Result Validation of Outsourced Machine Learning Workloads via TEE.
NSF RINGS Workshop, Jun 2023.

Academic Service

Program Committee Member and Reviewer. USENIX Security 2027, INFOCOM 2026